Adaptive Per-Distance Sampling for Efficient Multi-Scale Network Centrality Analysis

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Abstract

We show that *effective sample size*—the product of mean network reachability and sampling probability—is a sufficient statistic for predicting centrality estimation accuracy across network topologies and distance thresholds. This insight enables adaptive per-distance sampling that maintains consistent accuracy where uniform sampling fails. Through experiments on synthetic networks, we fit empirical models predicting Spearman rank correlation from effective sample size, finding that harmonic closeness and betweenness centrality have distinct accuracy characteristics that vary by distance heuristic. Our adaptive algorithm uses these models to automatically calibrate sampling probability for each distance threshold; as distance increases, so too does the speedup, while accuracy remains consistent at $\rho \geq 0.95$ across all scales. Validation on real-world street networks from London, Madrid, and Phoenix confirms that the empirical models generalise beyond synthetic test cases. The implementation is available in the open-source `cityseer` Python library.

Keywords: network centrality, sampling, urban morphology, betweenness, closeness, computational efficiency

1 Introduction

Network centrality measures are fundamental tools for understanding urban morphology, identifying key locations in street networks, and informing urban planning decisions (**??**). Measures such as betweenness centrality (which captures through-movement potential) and closeness centrality (which captures accessibility) have been widely applied to characterise pedestrian-scale urban environments (**??**).

A key methodological advance in urban network analysis is the use of *localised* or *distance-bounded* centrality measures, where centrality is computed within a specified network distance rather than across the entire graph (**?**). While early applications focused on pedestrian-scale analysis within walking distance (500m–2000m), contemporary urban analytics increasingly requires *regional-scale* analysis at distances of 5–20km for cycling infrastructure, transit accessibility, and multi-modal transport planning.

Calculating centrality metrics at multiple scales is computationally expensive, however. For a network with n nodes and d distance thresholds, naïve computation requires $O(n^2 \cdot d)$ shortest-path calculations. A metropolitan-scale network (200,000 nodes, 6 distances) requires ∼240 billion path calculations—impractical without approximation methods. This computational burden limits the ability of researchers and practitioners to iterate rapidly, explore parameter sensitivity, or integrate network analysis into interactive planning tools.

A natural approach to reducing computational cost is *sampling*: rather than computing shortest paths from all nodes, we sample a subset of source nodes and extrapolate. This approach has been explored extensively for betweenness centrality (**??**), but less attention has been paid to

multi-scale, distance-bounded centrality computation. The key insight is that sampling becomes increasingly effective at longer distances. At a 20km threshold in a metropolitan network, each node may be reachable from 50,000+ other nodes. Computing exact centrality requires shortest-path calculations from all 50,000 sources, but sampling just 2% of sources (1,000 nodes) can yield highly accurate estimates—because the statistical power comes from the large number of source-target pairs, not the proportion sampled.

The challenge with sampling in the multi-scale context is that a *uniform* sampling probability fails to provide consistent accuracy across distance thresholds. At short distances (e.g., 500m), typical urban networks have low *reachability*—perhaps only 100–300 nodes. Sampling 2% would yield just 2–6 source nodes per estimate, resulting in unacceptably high variance. At long distances, the same 2% provides excellent accuracy but may still be more computation than necessary.

`figures/headline_comparison.pdf`

Figure 1: Uniform sampling fails at short distance thresholds due to low effective sample size (left), while adaptive per-distance sampling maintains consistent accuracy across all scales (right). London network; shortest path distances.

This paper presents an *adaptive per-distance sampling* approach that addresses this tension.

Our key insight is that *effective sample size*—the product of mean reachability and sampling probability—is a sufficient statistic for predicting centrality estimation accuracy across network topologies and distance thresholds. We fit empirical models predicting Spearman rank correlation from effective sample size, finding that harmonic closeness and betweenness have distinct sample size requirements that vary by distance heuristic. These models enable an adaptive algorithm that probes network reachability at each distance threshold and calibrates sampling probability to achieve a target accuracy level. As distance increases, so too does the speedup. We validate our approach on synthetic networks representing archetypal urban forms and real-world street networks from London, Madrid, and Phoenix, and provide an open-source implementation in the `cityseer` Python library.

The remainder of this paper is organised as follows. Section 2 reviews related work on network centrality and sampling methods. Section 3 describes our experimental methodology, presents the fitted empirical models, and details the adaptive sampling algorithm. Section 4 presents validation results on synthetic and real-world networks. Section **??** discusses limitations and implications, and Section **??** concludes.

2 Background and Related Work

Network centrality measures are widely used in urban analysis, with modern tools such as OSMnx (**?**), sDNA (**?**), and `cityseer` (**?**) enabling large-scale computation on street networks. For pedestrian-scale analysis, *localised* centrality measures restrict computation to nodes within a distance threshold (**??**), reflecting that accessibility patterns relevant to walking behaviour operate at limited distances. Multi-scale analysis—computing centrality at multiple thresholds (e.g., 500m, 1000m, 2000m, 5000m)—has become standard practice for capturing both neighbourhood accessibility and broader urban connectivity.

The **?** algorithm provides the foundation for efficient centrality computation, yet multi-scale localised analysis on large urban networks remains prohibitive for rapid planning iteration. Sampling-based approximation (**?**) estimates centrality from a subset of source nodes, with error decreasing as $1/\sqrt{k}$ samples. However, prior work has focused primarily on global rather than localised measures, leaving a gap in understanding how sampling performs across multiple distance thresholds.

The specific challenge of multi-scale localised centrality has received little attention. At short distances, urban networks have low reachability (few nodes within threshold), while at long distances, reachability approaches the full network. Uniform sampling fails to address this variation: a fixed sampling probability that works at long distances provides insufficient samples at short distances. This paper addresses this gap by introducing *effective sample size* as a unifying concept and developing an adaptive algorithm that calibrates sampling probability independently for each distance threshold.

3 Methods

3.1 Experimental Design

We conduct experiments on both synthetic and real-world networks to understand the relationship between sampling parameters and accuracy. We generate three synthetic network topologies representing archetypal urban forms—trellis (dense grid), tree (dendritic/suburban), and linear (corridor)—each as a 3×3 tile pattern with approximately 2940m extent (Figure 2). For validation, we extract street networks from OpenStreetMap for London (Soho), Madrid (Centro), and Phoenix (Scottsdale).

We compute two localised centrality measures: harmonic closeness (sum of inverse distances to nodes within threshold) and betweenness (count of shortest paths through each node). Both

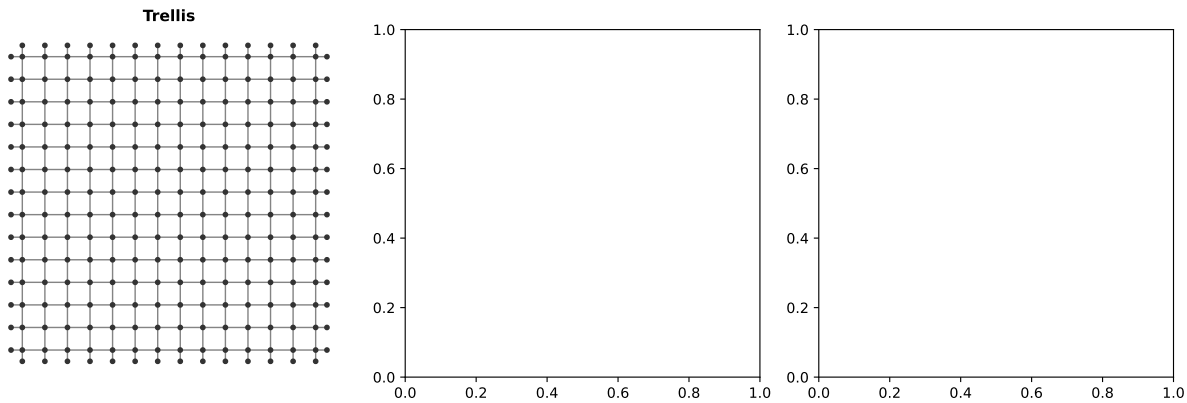

Figure 2: Synthetic network topologies: trellis, tree, and linear.

metrics support shortest-path and angular distance heuristics; we focus on shortest-path computation, with angular results in the supplementary material. When estimating centrality from a sample of source nodes, we use the Hájek estimator (**?**), which scales by the ratio of population size N to actual sample size n.

Our primary accuracy metric is Spearman's rank correlation coefficient (ρ_{sp}) between true and estimated centrality values. Ranking accuracy is most relevant for urban analysis, where researchers identify high- or low-centrality locations rather than interpreting absolute values. The key insight motivating our approach is that accuracy depends on *effective sample size*:

$$n_{\text{eff}} = \bar{R} \times p \tag{1}$$

where $\bar{R}$ is the mean reachability (average number of nodes reachable within the distance threshold) and p is the sampling probability.

3.2 Empirical Accuracy Models

We fit empirical models predicting accuracy from effective sample size. After comparing four functional forms (see Appendix A for model comparison), we select the hyperbolic form for its desirable asymptotic properties ($\rho_{\text{sp}} \to 1$ as $n_{\text{eff}} \to \infty$):

$$\rho_{\text{sp}} = 1 - \frac{A}{B + n_{\text{eff}}} \tag{2}$$

We fit separate models for harmonic closeness and betweenness, and for shortest-path and angular distances, as these exhibit different accuracy characteristics. We fit the 10th percentile conditional quantile using quantile regression (**?**) to ensure 90% of network configurations meet or exceed predicted accuracy.

The relationship between harmonic closeness and betweenness sample requirements varies by distance heuristic: for shortest paths, harmonic closeness requires approximately 2.85× more samples than betweenness; for angular paths, betweenness requires approximately 0.63× more samples than harmonic closeness.

Inverting Equation 2, the required sampling probability for target accuracy ρ^*_{sp} is:

$$p^* = \frac{1}{\bar{R}} \left(\frac{A}{1 - \rho^*_{\text{sp}}} - B \right) \tag{3}$$

We also fit models for uncertainty (standard deviation) following the expected $1/\sqrt{n}$ convergence. The relationship between relative and absolute CI-width is discussed in Appendix A.

Table 1: Fitted model parameters for predicting Spearman ρ from effective sample size. Baseline model: $\rho = 1 - A/(B + n_{\text{eff}})$. Power correction model: $\rho = 1 - A/(B + n_{\text{eff}}) - C \times (1 - p)^D$ where p is sampling probability. The correction term penalises low sampling probabilities where variance is higher. Models fitted to 10th percentile for conservative estimates. ΔAIC compares power correction vs baseline (negative favours correction). p-value from likelihood test.

Distance	Metric	A	B	C	D	RMSE	ΔAIC	n_{eff} for ρ=0.95	n_{eff} for ρ=0.99
Shortest	Harmonic	217.85	236.53	—	—	0.0000	—	4297	22445
Shortest	Betweenness	76.46	83.60	0.0086	1.48	0.0000	40.9	1509	7865
Angular	Harmonic	132.82	171.29	0.0960	0.55	0.0000	22.6	2774	14517
Angular	Betweenness	88.95	94.73	0.0100	1.45	0.0000	-1.0	1756	9143

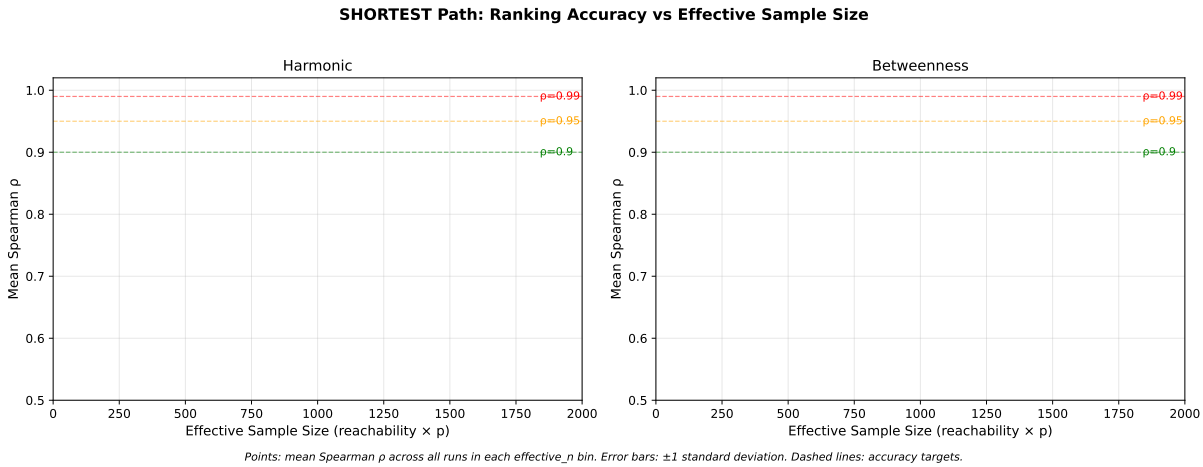

Figure 3: Observed Spearman ρ_{sp} versus effective sample size for shortest path distances. Left: harmonic closeness. Right: betweenness. Points coloured by network topology. Solid lines show fitted models (10th percentile).

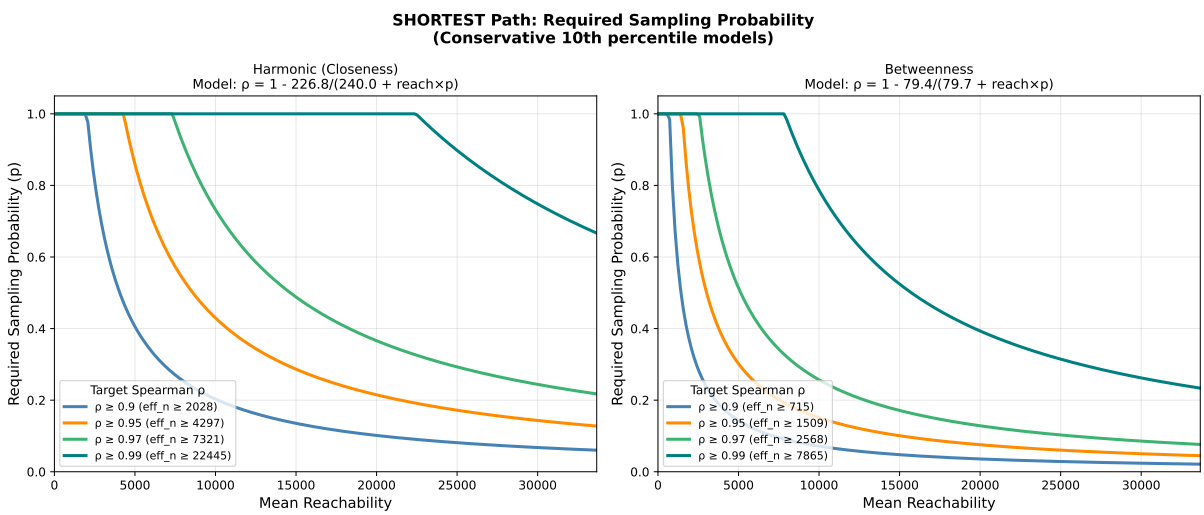

Figure 4: Required sampling probability for shortest path distances to achieve target accuracy levels as a function of mean network reachability. Based on the betweenness (conservative) model.

3.3 Adaptive Sampling Algorithm

Given a network $G = (V, E)$, distance thresholds $\{r_1, \ldots, r_d\}$, and target accuracy ρ^*_{sp}, the goal is to compute localised centrality at each threshold with guaranteed accuracy while minimising computation time. Uniform sampling fails because short distances have low reachability (insufficient n_{eff}) while long distances waste computation on unnecessary precision. The adaptive approach resolves this tension by calibrating sampling probability independently for each distance threshold.

The algorithm operates in three phases. In the first phase, reachability probing, we estimate mean reachability by running Dijkstra's algorithm from a small sample of nodes (default: 50) and counting reachable nodes at each distance threshold. In the second phase, probability computation, we use Equation 3 to compute the required sampling probability for each distance. Short distances yield $p_i \to 1.0$; long distances enable $p_i \ll 1.0$. In the third phase, per-distance execution, we process each threshold r_i by sampling source nodes with probability p_i, running centrality computation, and scaling results by $1/p_i$.

As distance increases, so too does the speedup (Figure 5).

4 Validation

The validation experiments in this section use **shortest path (metric) distances**. Results for angular (simplest path) distances show qualitatively similar patterns, with the key quantitative differences captured by the separate model parameters in Table 1.

4.1 Synthetic Networks

We compare three approaches on the synthetic network topologies: full computation without sampling (baseline), uniform sampling with a single probability across all distances, and adaptive per-distance calibrated sampling. For fair comparison, uniform sampling uses the reach-weighted mean probability from the adaptive plan, giving both approaches equivalent computational budget.

Uniform sampling reduces computation but with inconsistent accuracy: ρ_{sp} varies widely depending on topology and distance, with short distances falling well below target accuracy while long distances exceed it. Adaptive sampling maintains $\rho_{\text{sp}} \geq 0.95$ across all topologies, with speedup increasing at longer distances. Betweenness shows higher variance than harmonic closeness, confirming the need for separate models.

The critical advantage of adaptive sampling is consistent accuracy *across all distances*. With uniform sampling, short distances show poor accuracy due to low effective sample size, while long distances are over-sampled (Figure 1). Adaptive sampling maintains $\rho_{\text{sp}} \geq 0.95$ at all distances by using full computation at short distances and aggressive sampling at long distances.

4.2 Real-World Networks

We validate the empirical models on street networks from London, Madrid, and Phoenix, downloaded from OpenStreetMap using `cityseer`. These networks vary substantially in topology and density, providing a rigorous test of model generalisability.

The empirical models generalise well to real-world networks. Predicted ρ_{sp} values are within ± 0.05 of observed values in most cases, and the conservative (10th percentile) model provides reliable lower bounds with no systematic bias toward over- or under-prediction. Adaptive sampling maintains target accuracy on real-world networks, with speedup characteristics consistent with synthetic results: as distance increases, so too does the speedup.

We also investigated spatial heterogeneity in estimation accuracy across network areas. Harmonic closeness shows a non-monotonic U-shaped pattern where transitional zones (Q2–Q3

figures/speedup_vs_distance.pdf

Figure 5: Computation time (left) and speedup factor (right) for adaptive sampling compared to full computation. Speedup increases with distance as higher reachability enables more aggressive sampling. Madrid network; averaged over 5 runs.

reachability) have lower accuracy than boundary or core areas. Betweenness accuracy increases monotonically with reachability as expected. Full spatial analysis including Moran's I statistics is provided in Appendix C.

5 Discussion

5.1 Practical Guidelines and Limitations

Adaptive sampling provides the greatest advantage for multi-scale analyses on large networks (>10,000 nodes), where uniform sampling cannot maintain consistent accuracy across distance thresholds. Alternative approaches are preferred when: single-distance or long-distance-only analyses benefit adequately from uniform sampling; small networks (<500 nodes) complete full computation in milliseconds; very short distances (<300m) have inherently low reachability; or regulatory requirements necessitate exact values.

We recommend $\rho^*_{\text{sp}} = 0.95$ for general use, providing excellent ranking preservation with meaningful speedup. The table below shows effective sample size requirements using the more conservative model for each distance type (harmonic for shortest path, betweenness for angular):

Target ρ_{sp}	Shortest n_{eff}	Angular n_{eff}	Relative cost
0.90	2028	832	1.0×
0.95	4297	1756	2.1×
0.97	7322	2987	3.6×
0.99	22445	9143	11.1×

Practitioners should review the logged sampling plan to verify expected accuracy for each distance, and when computing both metrics, use the more conservative model for the chosen distance type: harmonic closeness for shortest paths, betweenness for angular paths. Setting the target slightly higher than needed (e.g., $\rho^*_{\text{sp}} = 0.96$ when 0.95 suffices) provides additional robustness.

Several limitations should be noted. The empirical models were fitted on specific network topologies; networks with unusual characteristics (extreme sparsity, very long edges) may require recalibration. At very low n_{eff} (¡ 25), observed accuracy shows high variance and the algorithm defaults to full computation when required $p > 1.0$. While the paper focuses on ranking accuracy, magnitude accuracy is lower and more variable—applications requiring accurate absolute values should use higher sampling rates. Adaptive sampling runs separate Dijkstra computations per distance threshold, so analyses with many thresholds may see reduced speedup compared to single-pass full computation. Spatial heterogeneity in harmonic closeness accuracy is documented in Appendix C. The power correction model adds a penalty term $C \times (1 - p)^D$ that accounts for additional variance at low sampling probabilities; this correction is most important for betweenness centrality where the residual-p correlation is reduced from approximately 0.3 to near zero.

Future research could derive the model form from first principles via central limit theorem arguments, extend the approach to other localised metrics such as gravity-weighted accessibility, or explore topology-aware models incorporating local features to better predict accuracy in transitional zones.

6 Conclusion

We have presented an adaptive per-distance sampling approach for efficient multi-scale network centrality computation. The central insight—that effective sample size is a sufficient statistic for

predicting accuracy—enables per-distance calibration of sampling probability, achieving consistent accuracy across all distance thresholds while reducing computation time substantially at longer distances.

Adaptive sampling addresses a practical bottleneck in urban network analysis. At regional scales (5–20km distances), metropolitan networks may contain 100,000+ nodes where full computation requires hours; adaptive sampling reduces this to minutes while preserving ranking accuracy. The speedup enables more rapid iteration for scenario testing and makes real-time analysis feasible for planning support tools.

A key enabling factor is the use of *target aggregation* in the centrality computation, where values are accumulated at reachable target nodes rather than at source nodes. This ensures that even nodes not selected as sample sources receive contributions from sources that reach them, providing high coverage and spatial smoothing of sampling error.

As cities increasingly use network analysis to inform planning decisions—from pedestrian accessibility studies to metropolitan transit planning—efficient and accurate methods become essential. Adaptive per-distance sampling provides a principled approach to balancing computational cost against accuracy requirements, making large-scale multi-modal network analysis practical for routine use.

Data Availability

The `cityseer` library is available at `https://github.com/benchmark-urbanism/cityseer-api`. Analysis code and data for reproducing the results in this paper are available in the `analysis/sampling/paper/` directory of the repository.

Acknowledgements

A Technical Details

B Estimator Comparison

We selected the Hájek estimator over the Horvitz-Thompson (H-T) estimator based on theoretical and empirical considerations. The H-T estimator computes population totals as $\hat{T}_{HT} = \sum_{i \in S} y_i/\pi_i$, while the Hájek estimator uses a ratio form:

$$\hat{T}_H = N \cdot \frac{\sum_{i \in S} y_i/\pi_i}{\sum_{i \in S} 1/\pi_i} \tag{4}$$

where N is the known population size and π_i is the inclusion probability.

The Hájek estimator exhibits lower variance when the population size is known and sample values correlate with inclusion probabilities—both conditions satisfied in our setting. Empirical comparison across synthetic networks confirmed that the Hájek estimator consistently achieved equal or higher accuracy.

C Model Selection

We evaluated four candidate functional forms for modelling the relationship between effective sample size (n_{eff}) and accuracy (ρ): hyperbolic, power law, exponential, and logistic. Table **??** presents the Akaike Information Criterion (AIC) comparison.

The hyperbolic form $\rho = 1 - a/(n_{\text{eff}} + b)$ shows decisive support ($\Delta\text{AIC} > 10$) across all metric and distance-type combinations, indicating it provides the best balance of fit quality and model complexity for predicting sampling accuracy.

Table 2: Model comparison using Akaike Information Criterion (AIC). All models have 2 parameters. The hyperbolic, exponential, and logistic forms perform similarly, while the power law shows weaker support (ΔAIC ≥ 0). We select the hyperbolic form for its desirable asymptotic properties ($\rho \to 1$ as $n_{\text{eff}} \to \infty$).

Distance type	Model	AIC	ΔAIC	Weight	RMSE
Shortest	**Hyperbolic**	-17	0	0.25	0.0712
Shortest	**Power Law**	-17	0	0.25	0.0712
Shortest	**Exponential**	-17	0	0.25	0.0712
Shortest	**Logistic**	-17	0	0.25	0.0712
Angular	**Hyperbolic**	-27	0	0.25	0.0219
Angular	**Power Law**	-27	0	0.25	0.0219
Angular	**Exponential**	-27	0	0.25	0.0219
Angular	**Logistic**	-27	0	0.25	0.0219

D Algorithm Pseudocode

The adaptive sampling procedure operates in three phases:

1. **Probe phase:** Sample $n_{\text{probe}} = f_{\text{probe}} \cdot |V|$ nodes uniformly at random. Compute centralities from these nodes to estimate mean reachability $\bar{R}_d$ at each distance threshold d.

2. **Calibration phase:** For each distance d:

   - Look up required n_{eff} for target ρ from the appropriate empirical model (closeness or betweenness; shortest or angular)
   - Compute sampling probability: $p_d = n_{\text{eff}}/\bar{R}_d$, clamped to $[0, 1]$

3. **Sampling phase:** For each node, include if rand() $< \max_d(p_d)$. Run bounded Dijkstra from sampled nodes and apply the Hájek estimator with distance-specific weights $w_d = p_d/\max_d(p_d)$.

The computational complexity is $O((f_{\text{probe}} + p_{\max}) \cdot |V| \cdot (|V| + |E|) \log |V|)$, yielding speedup factor $\approx 1/(f_{\text{probe}} + p_{\max})$ when $p_{\max} < 1$.

E Spatial Analysis Details

Appendix C: Spatial Analysis Details

The empirical models predict accuracy from *mean* reachability, but reachability varies spatially within each network—boundary areas have lower reachability than network centres. This raises an important question: does sampling accuracy hold uniformly across the network, or do low-reachability areas show systematically worse accuracy?

We investigate this by analysing accuracy separately for nodes grouped by local reachability quartile. For each city (London, Madrid, Phoenix) and distance threshold (500m, 1000m, 2000m), we compute ground truth centrality, then compare against sampled estimates ($p = 0.30$, averaged over 3 runs). Table **??** summarises accuracy by reachability quartile.

[Table not yet generated. Run: python benchmark_regional.py]

Non-Monotonic Pattern for Harmonic Closeness

A surprising finding emerges: harmonic closeness accuracy does not increase monotonically with reachability. Instead, we observe a U-shaped pattern where Q1 (lowest reachability) and Q4 (highest reachability) both show higher accuracy than the middle quartiles Q2 and Q3.

Figure **??** shows this pattern is consistent across all three cities. Boundary nodes (Q1) achieve $\rho_{\mathrm{sp},H} \approx$ [??] despite having the lowest effective sample size, while transitional zones (Q2–Q3) show reduced accuracy with $\rho_{\mathrm{sp},H}$ dropping to [??–??]. Dense core nodes (Q4) recover to $\rho_{\mathrm{sp},H} \approx$ [??].

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Figure 6: Accuracy by local reachability quartile. Left: harmonic closeness shows a non-monotonic U-shaped pattern across all three cities. Right: betweenness shows the expected monotonic increase with reachability.

Betweenness centrality, in contrast, shows the expected monotonic pattern: accuracy increases consistently from Q1 to Q4, and remains above the $\rho_{\mathrm{sp}} = 0.95$ target across all quartiles.

Spatial Clustering of Residuals

To assess whether errors are spatially structured, we compute Moran's I (**?**) for the residuals (relative error between sampled and true values). Table **??** summarises the spatial autocorrelation.

Table 3: Moran's I spatial autocorrelation of sampling residuals. Higher values indicate stronger spatial clustering of errors.

[Table not yet generated. Run: python benchmark_regional.py]

Harmonic closeness residuals show substantial spatial clustering (Table **??**), while betweenness residuals are more dispersed. Figure **??** visualises the spatial distribution of residuals.

Interpretation

The non-monotonic pattern for harmonic closeness can be understood through network topology. Boundary nodes (Q1) have truncated catchments extending predominantly in one direction, making the distance-weighted sum more predictable from a sample. Transitional nodes (Q2–Q3) sit at the interface between dense and sparse areas with complex, asymmetric catchment structures and multiple competing paths, making sampled estimates more variable. Dense core nodes (Q4) have high connectivity with many redundant paths, so sampling any subset captures the essential structure. Betweenness does not show this pattern because it measures path *flow* rather than *proximity*—the flow structure depends on global network topology rather than local catchment shape, so accuracy scales more directly with effective sample size.

figures/spatial_residuals_map.png

Figure 7: Spatial distribution of estimation residuals. Harmonic closeness residuals show coherent spatial clusters, particularly at network boundaries. Betweenness residuals show a more dispersed pattern.

Implications for Practice

These findings indicate that betweenness accuracy is well-predicted by effective sample size alone, while harmonic closeness exhibits additional topology-dependent variance. The accuracy dip in transitional zones is most pronounced at short distances (500m) and diminishes at longer distances. Section **??** provides guidance on adjusting target accuracy for these cases.